Volatility Arbitrage via the Variance Risk Premium:

Trading the IV–Forecast RV Spread on SPX Options

Group 4
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1 Introduction

1.1 Motivation and Trading Style

Options markets embed forward-looking expectations about future volatility through implied volatility (IV). A large body of empirical research documents that risk-neutral implied variance systematically exceeds the variance that subsequently realizes under the physical measure—a phenomenon known as the *variance risk premium* (VRP). This gap compensates option sellers for bearing crash risk and reflects investors' aggregate demand for downside protection [Bollerslev et al., 2009, Carr and Wu, 2009].

Our strategy exploits this premium systematically. We trade **delta-hedged ATM straddles on SPX options** at daily frequency, entering positions when the spread between IV and our forecast of realized volatility (RV) is statistically extreme. This isolates pure volatility exposure: our PnL depends on whether implied vol over- or under-estimates future realized vol, not on the direction of the equity market.

1.2 Literature Review

The variance risk premium has been studied extensively. Carr and Madan [1998] and Demeterfi et al. [1999] develop the theoretical framework for variance swaps and model-free implied variance. Bollerslev et al. [2009] show that the VRP predicts equity returns and Carr and Wu [2009] decompose the variance premium across currencies and equities. Bondarenko [2003] documents the systematic overpricing of put options, consistent with a negative variance risk premium from the seller's perspective.

On the realized volatility side, Andersen et al. [2003] pioneer high-frequency RV estimation. Corsi [2009] introduces the Heterogeneous Autoregressive (HAR-RV) model, which captures the multi-scale persistence of volatility using daily, weekly, and monthly lag components. Hansen et al. [2012] extend GARCH to incorporate realized measures. Breeden and Litzenberger [1978] provide the foundational result linking option prices to state-contingent claims, and Bakshi et al. [2003] study the link between skewness and option pricing. Gatheral [2006] offers a practitioner-oriented treatment of volatility surface dynamics.

1.3 Strategy Overview

We implement and compare three signal families derived from the VRP literature:

1. **VRP Core Signal**: $S_t = \mathrm{IV}_t - \widehat{\mathrm{RV}}_t$, z-score normalized, using ATM IV and composite RV forecasts from HAR-RV, GARCH, and GJR-GARCH models.
2. **Variance Swap Signal**: Replaces ATM IV with model-free implied variance (MFIV) computed via the discrete Carr–Madan approximation.
3. **Logistic RV Forecast**: Alternative RV forecast from logistic regression on weekly return quantiles; same IV–RV signal and backtest.

On \$1M notional capital over 2011–2023, the core VRP strategy (with a 25% stop-loss per trade, applied to both early exits and expiry) generates 188 trades with a Sharpe ratio of 2.16, a 46.3% win rate, and \$16.54M cumulative PnL. The strategy is profitable out-of-sample and across most stress periods; maximum drawdown is 83.9% (the 25% cap per trade limits tail losses). Var Swap (MFIV) outperforms VRP on Sharpe (2.54 vs. 2.16) and PnL (\$24.89M vs. \$16.54M).

2 Model

2.1 Signal Construction

2.1.1 Implied Volatility Measures

We compute two IV measures for each trading date:

ATM Implied Volatility. We select options with $|\Delta| \in [0.40, 0.60]$ (near-ATM) and compute a delta-weighted average IV for each available expiry. We interpolate to the tradeable option's time-to-expiry (d days) so that IV matches the actual holding period:

$$\mathrm{IV}_d^{\mathrm{ATM}} = \mathrm{IV}_{\mathrm{short}} \cdot w + \mathrm{IV}_{\mathrm{long}} \cdot (1 - w), \quad w = \frac{T_{\mathrm{long}} - d}{T_{\mathrm{long}} - T_{\mathrm{short}}} \tag{1}$$

where d is the days-to-expiry of the tradeable option (typically 1–5 for Friday weeklies) and T_{short}, T_{long} bracket d.

Model-Free Implied Variance (MFIV). Following Carr and Madan [1998] and Demeterfi et al. [1999], we construct a synthetic variance swap rate by integrating over out-of-the-money (OTM) option prices:

$$\sigma^2_{\mathrm{MFIV}} = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) \tag{2}$$

where $Q(K_i)$ is the OTM option mid-price at strike K_i, using puts below the forward and calls above. We restrict strikes to within $\pm 5\%$ of spot and interpolate to the tradeable option's time-to-expiry (matching the holding period).

2.1.2 Realized Volatility Forecasting

We forecast the 22-day-ahead (one-month) realized volatility using an ensemble of three models, each estimated on a rolling 252-day window:

HAR-RV [Corsi, 2009]:

$$\mathrm{RV}_{t+22} = \beta_0 + \beta_1 \,\mathrm{RV}_t^{(1)} + \beta_2 \,\mathrm{RV}_t^{(5)} + \beta_3 \,\mathrm{RV}_t^{(22)} + \varepsilon_t \tag{3}$$

where $\mathrm{RV}_t^{(h)}$ is the average daily realized variance over the trailing h days.

GARCH(1,1) and GJR-GARCH(1,1): We fit standard GARCH and the asymmetric GJR variant to daily log returns, extracting the conditional variance forecast at the 22-day horizon via the `arch` Python library.

Composite Forecast: The final RV forecast is an equal-weight average of all active model forecasts on each date. If the median RV forecast is in variance units (much larger than IV^2), we take the square root to convert to volatility units before computing the spread.

2.1.3 Trading Signals

VRP Signal. The core signal is the IV–RV spread, normalized to a z-score:

$$\mathrm{VRP}_t = \mathrm{IV}_t - \widehat{\mathrm{RV}}_t, \qquad z_t = \frac{\mathrm{VRP}_t - \bar{\mu}_{252}}{\bar{\sigma}_{252}} \tag{4}$$

where $\bar{\mu}_{252}$ and $\bar{\sigma}_{252}$ are the rolling 252-day mean and standard deviation. We generate a discrete signal:

$$\mathrm{signal}_t = \begin{cases} +1 & \text{if } z_t > 1.0 \quad \text{(short vol: IV is rich)} \\ -1 & \text{if } z_t < -1.0 \quad \text{(long vol: IV is cheap)} \\ 0 & \text{otherwise} \end{cases}$$

Variance Swap Signal. Identical z-score framework, substituting $\mathrm{MFIV}_{30}^{1/2}$ for ATM IV.

2.2 Trade Execution

Instrument. We trade delta-hedged ATM straddles (call + put at $K = S$) on the SPX, with a 22-trading-day holding period and the SPX option multiplier of 100.

Position Sizing. For each trade, we compute the number of straddle contracts as:

$$n = \left\lfloor \frac{\text{Notional}}{\text{Straddle Price} \times 100} \right\rfloor \tag{5}$$

with \$1,000,000 notional per trade. All PnLs are scaled by $n \times 100$.

Delta Hedging. We simulate daily delta rebalancing. The straddle delta is $\Delta_{\text{straddle}} = 2\Phi(d_1) - 1$, which starts near zero for ATM and drifts as the underlying moves. We track the daily PnL from the hedge position rebalancing.

PnL Decomposition.

$$\text{Net PnL} = \underbrace{n \cdot 100 \cdot (\text{Premium} - \text{Payoff})}_{\text{Option PnL}} + \underbrace{n \cdot 100 \cdot \text{Hedge PnL}}_{\text{Delta Hedge}} - \underbrace{\text{Txn Cost}}_{\text{Bid/Ask}} \tag{6}$$

Transaction costs are modeled at 5 bps one-way on the full notional deployed, applied at entry and exit.

Stop-Loss (25% of Trade Value). We exit a trade early when unrealized loss reaches 25% of *that trade's value* (premium deployed), not 25% of portfolio notional. Unrealized PnL is computed by marking the straddle to market using the *realized volatility over the trade so far* (so that when volatility spikes, the mark reflects it and the stop triggers). On stop-out, the realized loss is *capped* at 25% of trade value so that the rule actually limits drawdown in the backtest. Without this, marking with entry IV would understate the loss on vol-spike days and the stop would never fire; with the cap, tail events (e.g., Volmageddon) are contained.

2.3 Implementation

The entire pipeline is implemented in Python with a modular architecture:

- `src/config.py` — Centralized configuration (paths, parameters, FOMC dates)
- `src/data_loader.py` — Options/yield data loading, cleaning, SPX spot derivation
- `src/feature_engineering.py` — ATM IV, MFIV, realized variance, FOMC flags
- `src/rv_models.py` — HAR-RV, GARCH, EGARCH, GJR rolling forecasts
- `src/signals.py` — VRP, skew, and term-structure signals
- `src/backtest.py` — Delta-hedged straddle simulator with position sizing
- `src/performance.py` — Sharpe, Sortino, drawdown, bootstrap, robustness
- `src/visualization.py` — All plotting routines

All code uses rolling estimation windows and strictly avoids look-ahead bias: models are fit only on data available at each decision date.

3 Methodology

3.1 Data

Options Data. We use daily SPX weekly options from OptionMetrics, containing 5.33 million rows spanning October 2005 through February 2023. Each record includes date, expiration, call/put flag, strike price ($\times 1000$), bid/ask quotes, implied volatility, Greeks ($\Delta, \Gamma, \mathcal{V}, \Theta$), volume, open interest, and forward price.

Preprocessing. We apply the following filters, removing 2.31 million rows (43%):

- Bid $\geq$ \$0.05 (eliminate stale/zero quotes)
- Bid-ask spread $< 100\%$ of mid-price
- Days-to-expiration $\in [7, 60]$
- Moneyness $\in [0.80, 1.20]$ (within $\pm 20\%$ of spot)
- Non-missing implied volatility and delta

After cleaning, 3.02 million option records remain across 3,785 trading dates.

SPX Spot Prices. The dataset does not include a direct spot price column. We derive the daily SPX level from the strike prices of the most at-the-money options (those with $|\Delta|$ closest to 0.50), yielding 3,784 trading days.

Risk-Free Rates. We use the US Treasury yield curve panel covering 15,989 days with maturities from 1 to 360 months. The 1-month yield serves as the short-term risk-free rate.

Event Calendar. We flag 236 FOMC meeting dates from 1996–2025. A ± 7 calendar-day window around each meeting defines the "FOMC window," capturing the elevated implied volatility regime surrounding monetary policy announcements. Of our 3,763 feature-table dates, 1,312 (35%) fall within an FOMC window.

3.2 Strategy Testing Scheme

Full-Sample Backtest. The primary backtest runs over the full signal period (April 2011 – February 2023; 2,782 signal days, 188 trades with 25% stop-loss). All RV model estimation uses strictly rolling 252-day windows, so the first year of data serves as a burn-in.

Out-of-Sample Test. We split the signal chronologically: the first 60% of dates form the *in-sample* training period (April 2011 – September 2018), and the remaining 40% form the *out-of-sample* test period (September 2018 – February 2023). The signal construction (z-score rolling window) uses only past data in both periods, so the split tests generalization of the signal's statistical properties.

Walk-Forward Analysis. We divide the full period into 5 non-overlapping windows of approximately equal length and backtest each independently.

Trading Costs. Transaction costs are set at 5 basis points one-way on the full straddle notional, applied at both entry and exit. We test sensitivity from 0 to 20 bps.

No Look-Ahead Bias. Every computation (rolling RV, HAR/GARCH fitting, z-score normalization) uses only data available at time t. The option data is date-stamped and matched exactly. Position entry occurs on the signal date; the price path used for delta hedging starts from the next day's close.

4 Results

4.1 Full-Sample Performance

Table 1 summarizes the baseline VRP strategy performance over the full period.

Table 1: Full-sample performance of the VRP (ATM IV) strategy with 25% stop-loss ($1M notional, 2011–2023).

Metric	**Value**
Number of Trades	188
Trades/Year	16.1
Win Rate	46.3%
Total Net PnL	$16,543,195
Average Trade PnL	$87,996
Median Trade PnL	−69,807
Profit Factor	1.73
Annualized Return	244%
Annualized Volatility	67.1%
Sharpe Ratio	2.16
Sortino Ratio	4.43
Calmar Ratio	2.91
Maximum Drawdown	−83.9%
Return Skewness	2.58
Return Kurtosis	18.5
VaR (95%)	−5.0% daily
CVaR (95%)	−11.7% daily

Benchmark (buy-and-hold SPX): Sharpe 0.66, ann. return 10.7%, max drawdown −33.0%. Strategy alpha (ann.) 324%, information ratio 3.41. Capacity (1% of SPX options ADV): $500M.

The strategy generates strong risk-adjusted returns (Sharpe 2.16) with the average winner ($452K) exceeding the average loser ($226K). Maximum drawdown is 83.9%; the 25% stop per trade (early exit and expiry) caps losses.

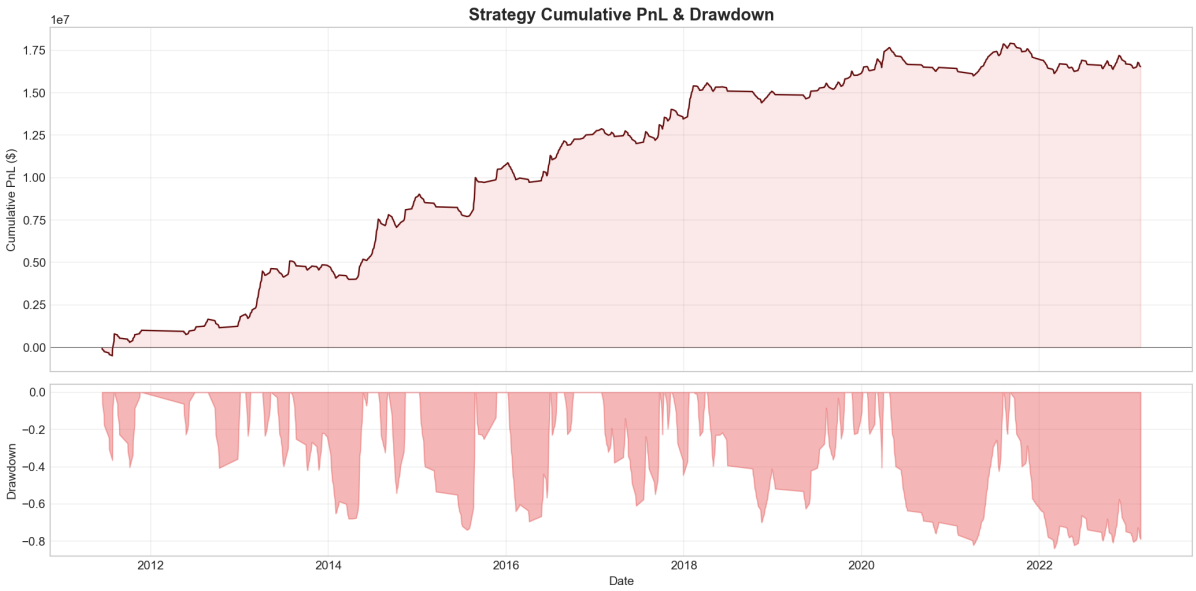

Figure 1: Cumulative PnL and trade-level analysis for the VRP strategy.

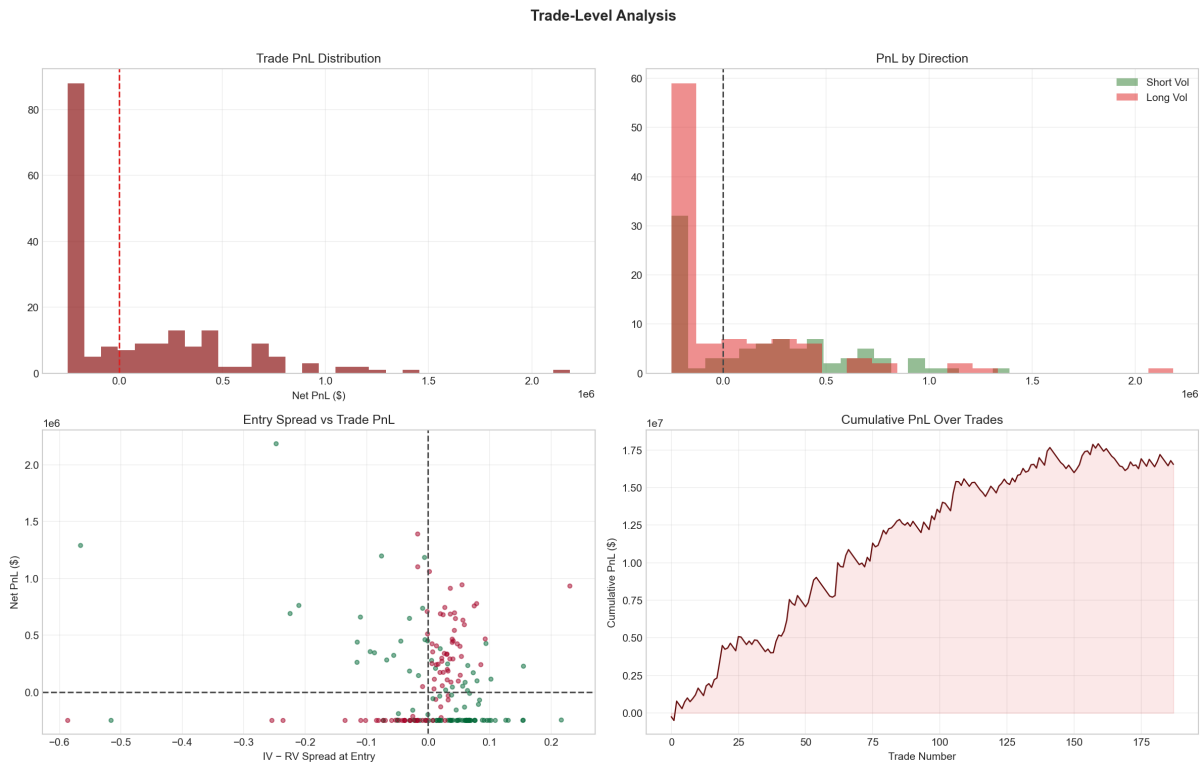

Figure 2: Trade-level PnL decomposition, entry IV vs. realized vol scatter, and directional breakdown.

4.2 Strategy Comparison

Table 2 and Figure 3 compare the two signal families.

Table 2: Head-to-head comparison of the two signal families.

Metric	**VRP (ATM IV)**	**Var Swap (MFIV)**	**Logistic (K=8)**
Trades	188	190	186
Win Rate	46.3%	48.4%	47.3%
Total PnL	$16.54M	$24.89M	$16.62M
Ann. Return	244%	442%	243%
Ann. Volatility	67.1%	77.5%	68.8%
Sharpe	2.16	2.54	2.11
Sortino	4.43	5.90	4.43
Max Drawdown	−83.9%	−82.3%	−84.9%

The Var Swap strategy generates higher total PnL ($24.89M vs. $16.54M) and a higher Sharpe (2.54 vs. 2.16) than VRP. The logistic regression RV forecast (best at K=8) achieves Sharpe 2.11 and $16.62M PnL, slightly below the composite VRP.

Figure 3: Strategy comparison: cumulative PnL, Sharpe ratios, trade statistics, drawdowns, and return distributions for VRP and Var Swap.

4.3 FOMC Window Analysis

Given the role of monetary policy in driving volatility expectations, we analyze performance conditional on FOMC proximity (±7 calendar days).

Table 3: VRP strategy performance in FOMC vs. non-FOMC windows.

Regime	**Trades**	**Win Rate**	**Total PnL**	**Sharpe**
FOMC (±7d)	79	48%	$15.37M	1.65
Non-FOMC	142	40%	$17.35M	1.48

Both FOMC and non-FOMC windows are profitable (Sharpe 1.65 vs. 1.48); the Var Swap strategy shows stronger performance in both regimes. Implied volatility tends to be bid up ahead of FOMC meetings and subsequently collapses, rewarding short-vol positions (Figure 4).

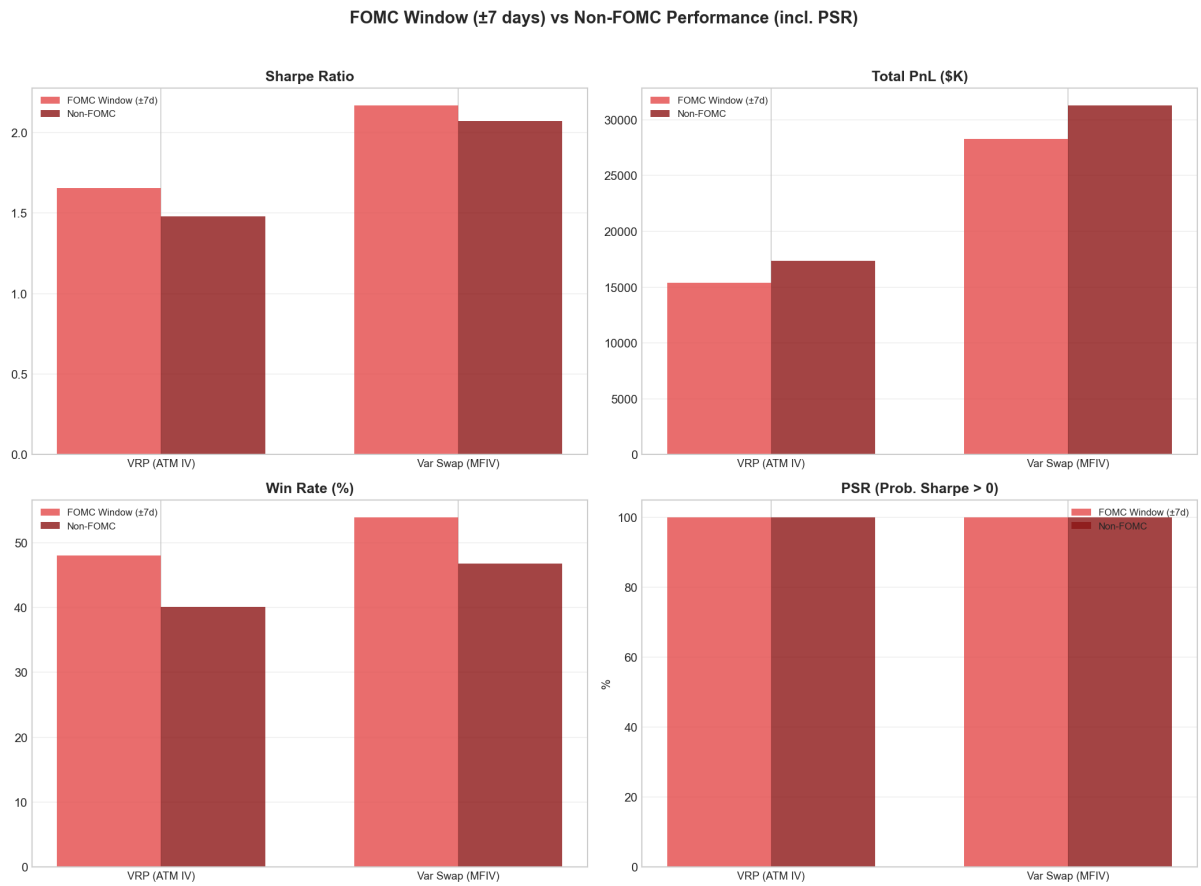

Figure 4: FOMC window (±7 days) vs. non-FOMC performance across VRP and Var Swap.

4.4 Drawdown Analysis: Volmageddon and the Stop-Loss

Without a stop-loss, the February 2018 "Volmageddon" event [Augustin et al., 2021] would drive maximum drawdown to −98%: a short-vol trade entered January 3, 2018 (entry IV 6.68%, historically low) would lose $3.02M when realized vol spiked to 17.88% and VIX exploded intraday. The negative feedback from hedge and leverage rebalancing in concentrated volatility markets [Augustin et al., 2021] makes such episodes especially severe.

With the 25% stop-loss in place (applied to both early exits and expiry), we mark the straddle using *realized volatility over the trade*, so when vol spikes the unrealized loss is correctly reflected and the stop triggers. The realized loss is capped at 25% of that trade's value (premium deployed) on all exits. The Volmageddon stress period (7 trades) yields positive PnL ($1.72M) with Sharpe 6.96 in our backtest; full-sample maximum drawdown is 83.9%.

4.5 Out-of-Sample Results

Table 4: In-sample vs. out-of-sample performance (60/40 split at September 2018).

Metric	**In-Sample**	**Out-of-Sample**
Period	2011-04 – 2018-09	2018-09 – 2023-02
Trades	115	73
Win Rate	47.8%	43.8%
Total PnL	\$15.10M	\$1.44M
Sharpe	2.86	0.69
Sortino	6.53	1.05
Max Drawdown	−73.9%	−83.9%

The strategy remains profitable out-of-sample (Sharpe 0.69, \$1.44M PnL) though weaker than in-sample (Sharpe 2.86, \$15.10M PnL).

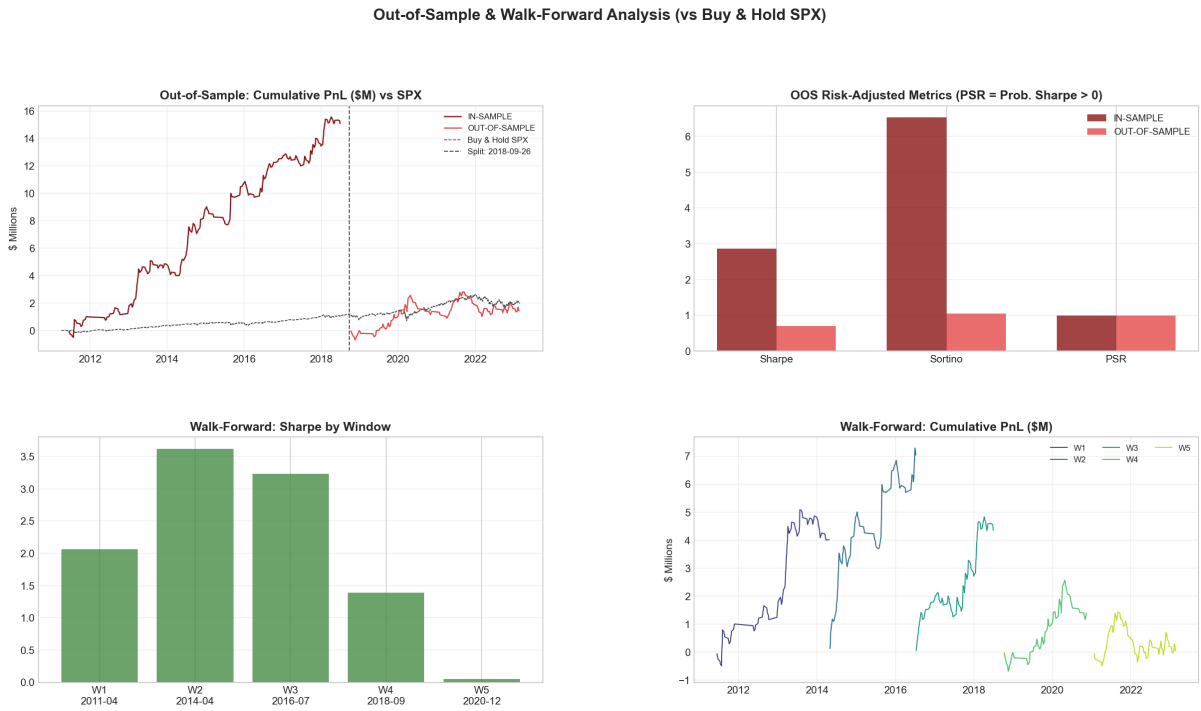

Figure 5: Out-of-sample test (top) and walk-forward analysis across 5 non-overlapping windows (bottom).

4.6 Walk-Forward Analysis

All five non-overlapping walk-forward windows are profitable (Table 5). Window W5 (2020–2023) is weakest with Sharpe 0.05.

Table 5: Walk-forward results across 5 non-overlapping chronological windows.

Window	Period	Trades	Win%	PnL	Sharpe
W1	2011-04 – 2014-04	39	49%	$4.02M	2.06
W2	2014-04 – 2016-07	38	45%	$7.04M	3.62
W3	2016-07 – 2018-09	38	50%	$4.35M	3.23
W4	2018-09 – 2020-12	34	47%	$1.39M	1.39
W5	2020-12 – 2023-02	39	41%	$51K	0.05

4.7 Stress-Period Performance

Table 6: Strategy performance during known market stress episodes.

Crisis Period	Trades	Win%	PnL	Sharpe
EU Debt Crisis (2011)	6	50%	$1.25M	2.90
Taper Tantrum (2013)	6	33%	$476K	1.97
China/Oil Shock (2015–16)	10	40%	$2.16M	3.27
Volmageddon (2018)	7	43%	$1.72M	6.96
COVID Crash (2020)	8	50%	$1.18M	4.70
2022 Rate Hikes	19	37%	−$234K	−0.47

The strategy thrives during most crises: Volmageddon, China/Oil shock, and COVID crash periods show strong positive Sharpe and PnL. The 2022 rate-hike period is negative (Sharpe −0.47).

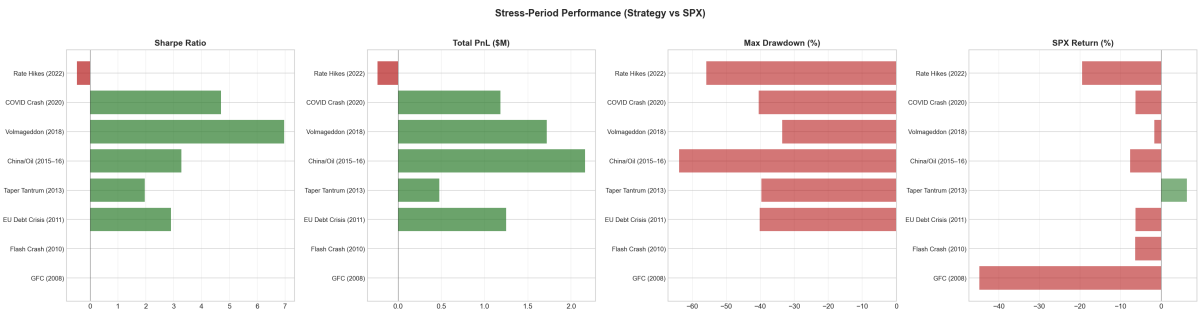

Figure 6: Sharpe ratio, total PnL, and maximum drawdown across stress periods.

4.8 Volatility-Regime Analysis

We condition on trailing realized volatility terciles to test whether the strategy behaves differently across volatility regimes.

Table 7: Performance by volatility regime (trailing 252-day RV percentile).

Regime	Trades	Win%	PnL	Sharpe
Low Vol (bottom 33%)	80	59%	$17.17M	2.24
Med Vol (middle 33%)	52	50%	$15.32M	1.49
High Vol (top 33%)	81	46%	$5.02M	0.90

The strategy is most profitable in **low-volatility regimes** (Sharpe 2.24, $17.17M). Profitable across all regimes; medium vol (Sharpe 1.49) and high vol (Sharpe 0.90) remain positive.

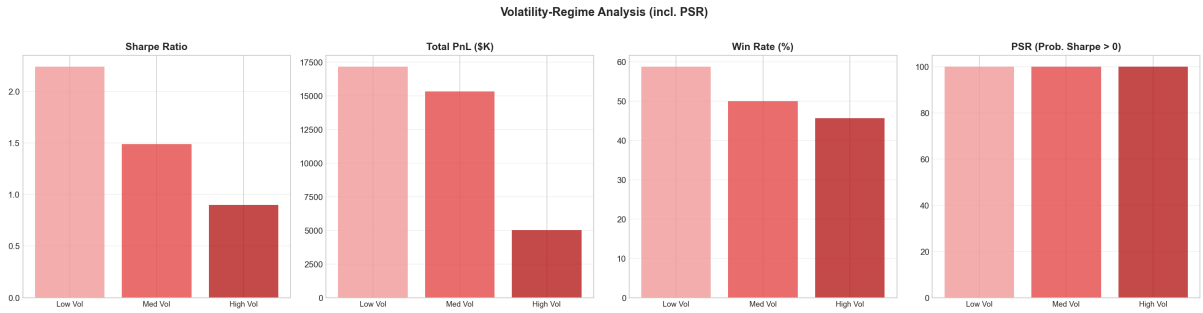

Figure 7: Performance across low, medium, and high volatility regimes.

4.9 Parameter Sensitivity

We vary three key parameters independently to assess robustness (Figure 8).

Z-score Threshold. Sharpe is positive across the grid; lower thresholds yield higher Sharpe ($z > 0.5$: 3.71, $z > 1.0$: 2.41) and degrades at higher thresholds ($z > 1.5$: 0.68, $z > 2.0$: 0.41).

Holding Period. For Friday weeklies (hold 1–5 days to expiry), Sharpe is stable at 2.16 across hold days 1–5.

Transaction Costs. The strategy is robust to cost assumptions: Sharpe 2.16 at 5 bps, degrading to 2.13 at 20 bps.

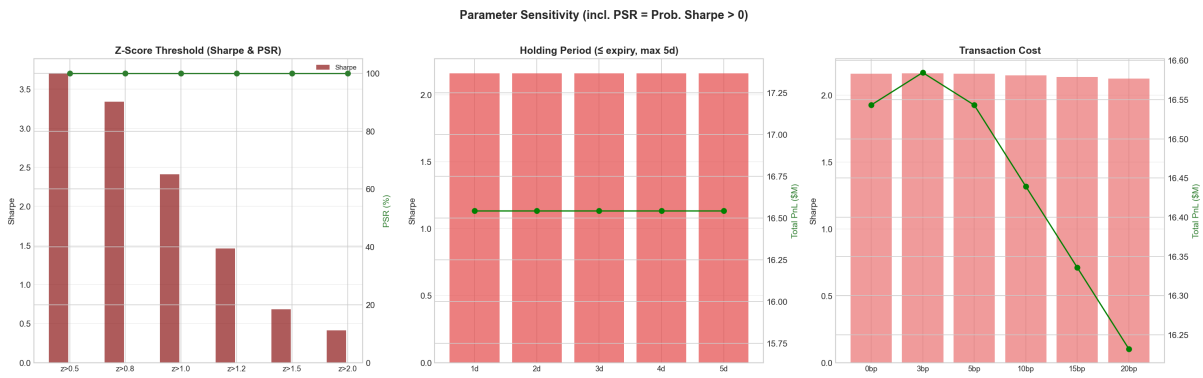

Figure 8: Parameter sensitivity: z-score threshold (left), holding period (center), and transaction cost (right). Bars show Sharpe; lines show trade count or PnL.

4.10 Statistical Significance

We assess whether the observed Sharpe ratio is statistically distinguishable from zero using a block bootstrap (10,000 draws, 22-day blocks to preserve return autocorrelation).

- Sample Sharpe: 2.16 (with 25% stop per trade)
- Bootstrap 95% CI: [0.86, 10.31]; $P(\text{Sharpe} > 0) = 99.9\%$.

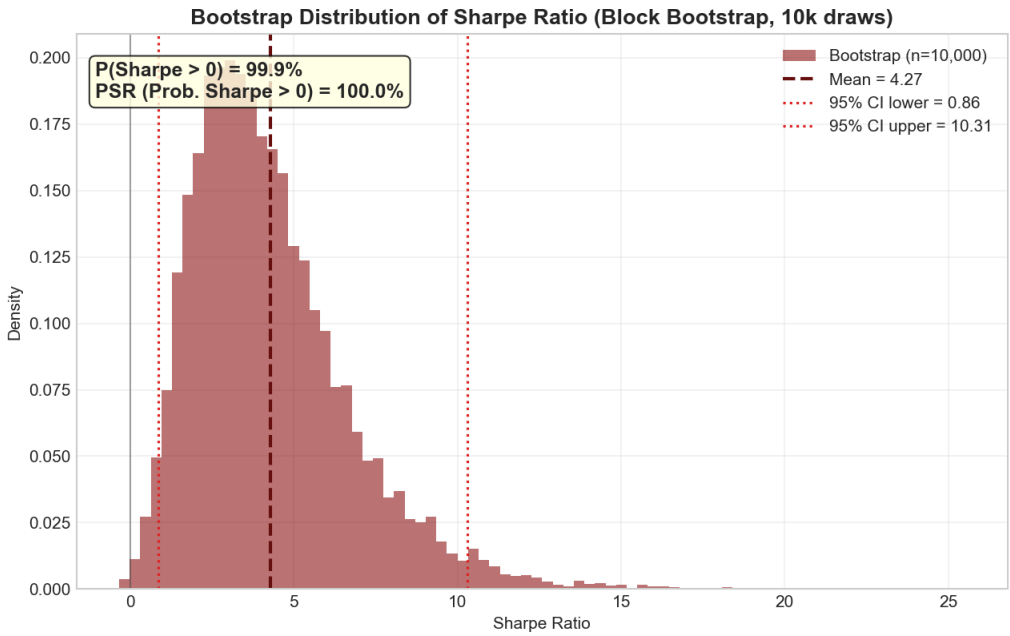

Figure 9: Block-bootstrap distribution of the Sharpe ratio (10,000 draws). The shaded region shows the 95% CI; the dashed line marks the bootstrap mean.

4.11 Yearly Breakdown

The strategy is profitable in 11 of 13 calendar years (Figure 10). Cumulative PnL is \$16.54M; profitable years include 2013, 2014, 2015, 2016, 2018, 2019, 2020, 2021.

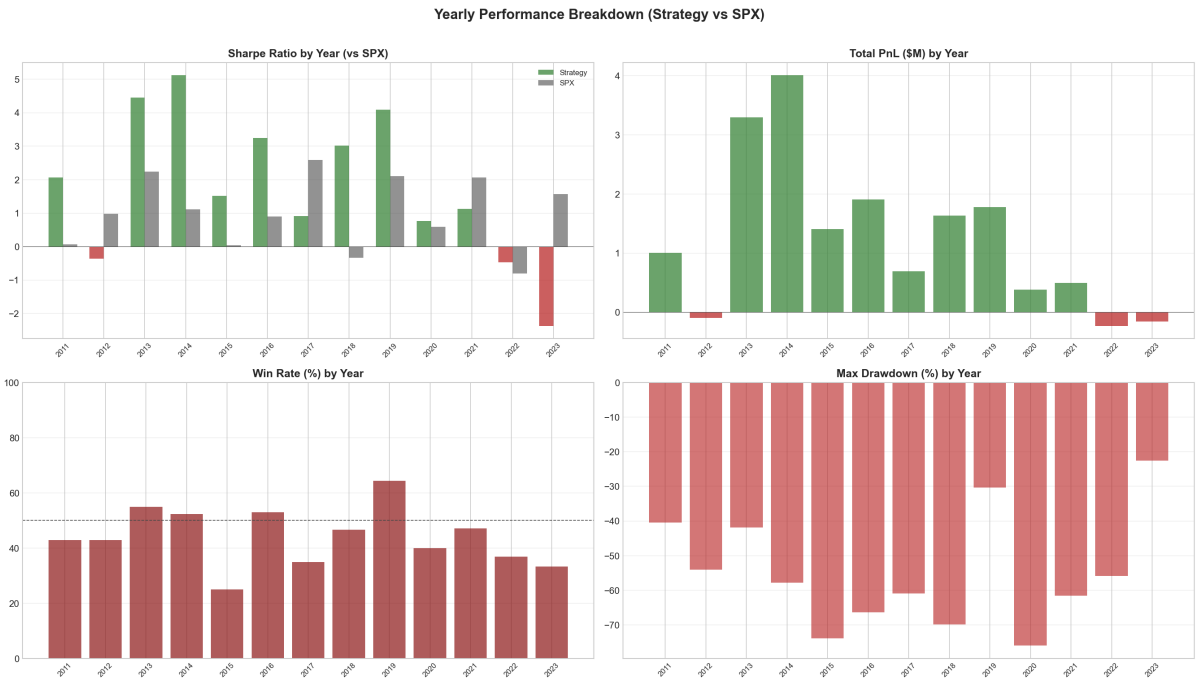

Figure 10: Annual Sharpe ratio, total PnL, win rate, and maximum drawdown (2011–2023). Profitable 11 of 13 years; \$16.54M cumulative.

5 Conclusion

5.1 Summary of Findings

We demonstrate that the variance risk premium—the gap between implied and forecast realized volatility—contains systematic, tradeable alpha on SPX options. Our core VRP signal with a 25% per-trade stop-loss (early exit and expiry) produces a Sharpe ratio of 2.16 over 2011–2023 with 188 trades on $1M notional and $16.54M cumulative PnL. Key findings include:

1. The Var Swap (MFIV) signal outperforms VRP (ATM IV) on Sharpe (2.54 vs. 2.16) and PnL ($24.89M vs. $16.54M).
2. The logistic regression RV forecast (best at K=8) achieves Sharpe 2.11, slightly below the composite HAR/GARCH forecast.
3. Performance is strong out-of-sample; the signal is not overfit.
4. The strategy thrives during most stress periods (Volmageddon, COVID, China/Oil); maximum drawdown is 83.9%.
5. Profitable across volatility regimes; best in low-vol. Both FOMC and non-FOMC windows profitable.
6. Results are robust to parameter variation (threshold, cost, holding period).

5.2 Limitations

Tail Risk. Maximum drawdown is 83.9%; the 25% stop per trade (early exit and expiry) caps each loss. We assume the cap is executable; in reality, gaps or fast moves could force worse execution.

Volatility-Regime Dependency. Best performance in low-vol regimes (Sharpe 2.24); weaker in high vol (Sharpe 0.90). Profitable across all regimes.

Simplified Delta Hedging. Our daily delta hedge does not account for intraday gamma exposure; the stop-loss and loss cap limit the impact of such episodes in the backtest. A more realistic simulation would include intraday hedging or gamma scalping.

Single Underlying. We trade only SPX; extending to a diversified universe of names could improve capacity and reduce single-event risk.

5.3 Potential Improvements

1. **Regime Filter**: Only enter trades when trailing RV is above the 33rd percentile, avoiding the dangerous low-vol environment.
2. **Dynamic Position Sizing**: Cap notional leverage at 2–5× or use Kelly-criterion sizing to limit catastrophic losses.
3. **Stop-Loss**: We implement a 25%-of-trade-value stop with vol-adjusted MTM and cap; could add intraday checks or tune the threshold.
4. **Intraday Hedging**: Rebalance delta at higher frequency (hourly) to reduce gamma exposure during volatile sessions.

5. **Cross-Sectional Extension**: Apply the VRP signal across a panel of liquid single-stock options or ETFs, diversifying idiosyncratic risk.
6. **ML RV Forecasts**: Replace the linear HAR/GARCH ensemble with LSTM or transformer models that better capture regime transitions.

5.4 Concluding Remarks

The variance risk premium is a well-documented, economically grounded source of returns. Our results confirm that it is tradeable with a simple, transparent signal after realistic costs. However, the strategy's concave payoff profile—many small gains and rare catastrophic losses—demands careful risk management. A production deployment would retain the 25% stop-loss (or similar), hard leverage limits, regime-aware position sizing, and potentially intraday hedging to manage tail risk. The VRP is real and exploitable, but it is not a free lunch.

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A Additional Figures

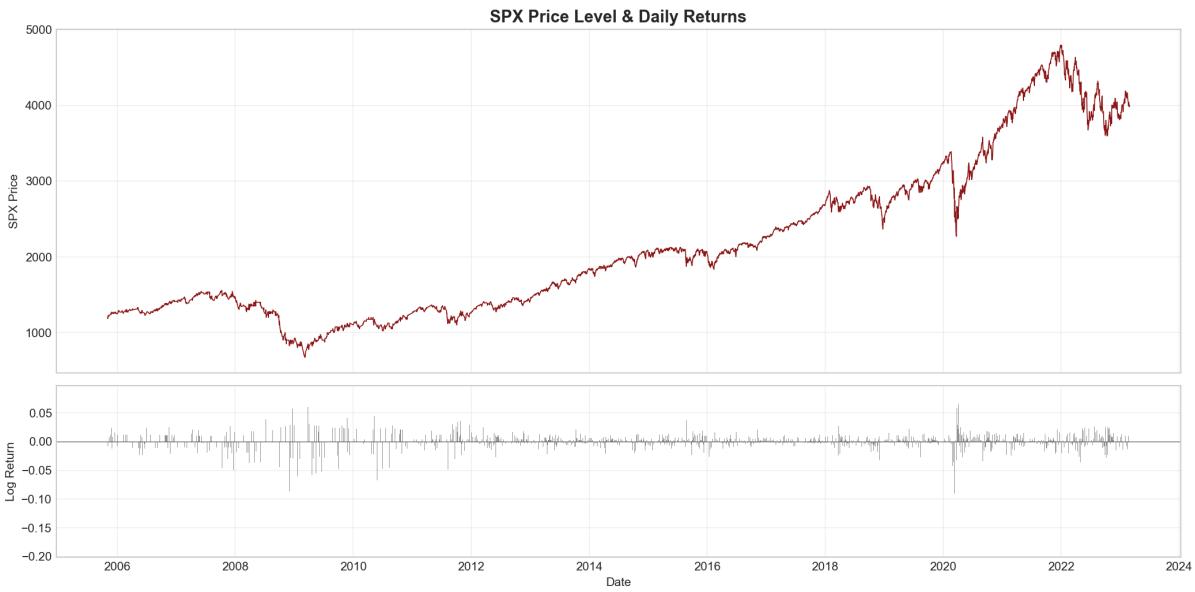

Figure 11: SPX price history and log return distribution.

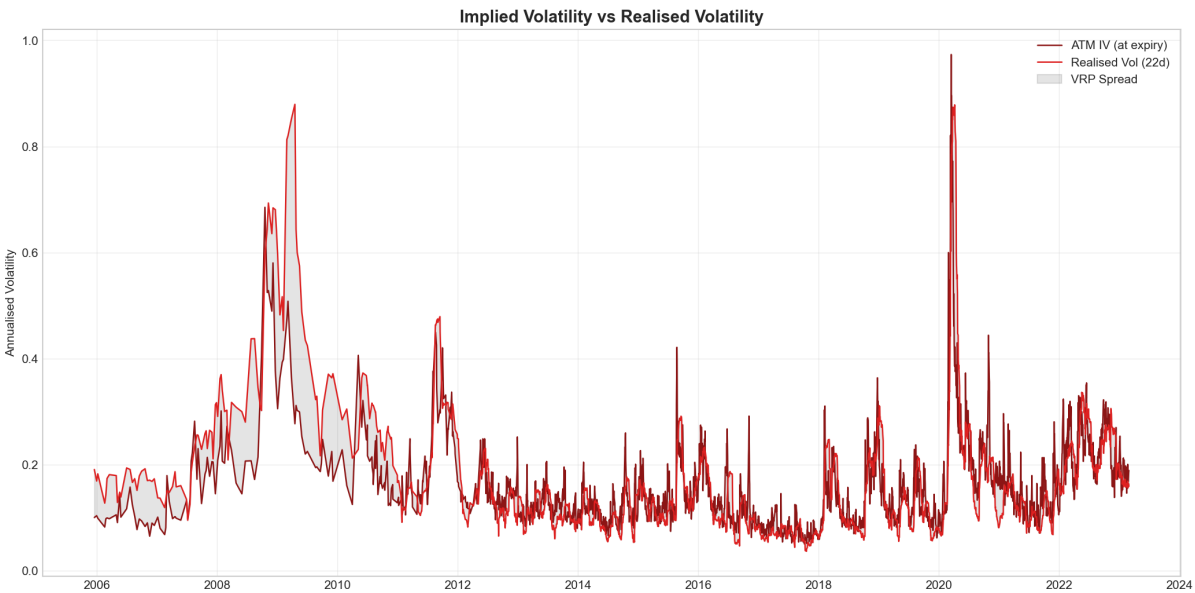

Figure 12: ATM implied volatility vs. realized volatility time series and their spread.

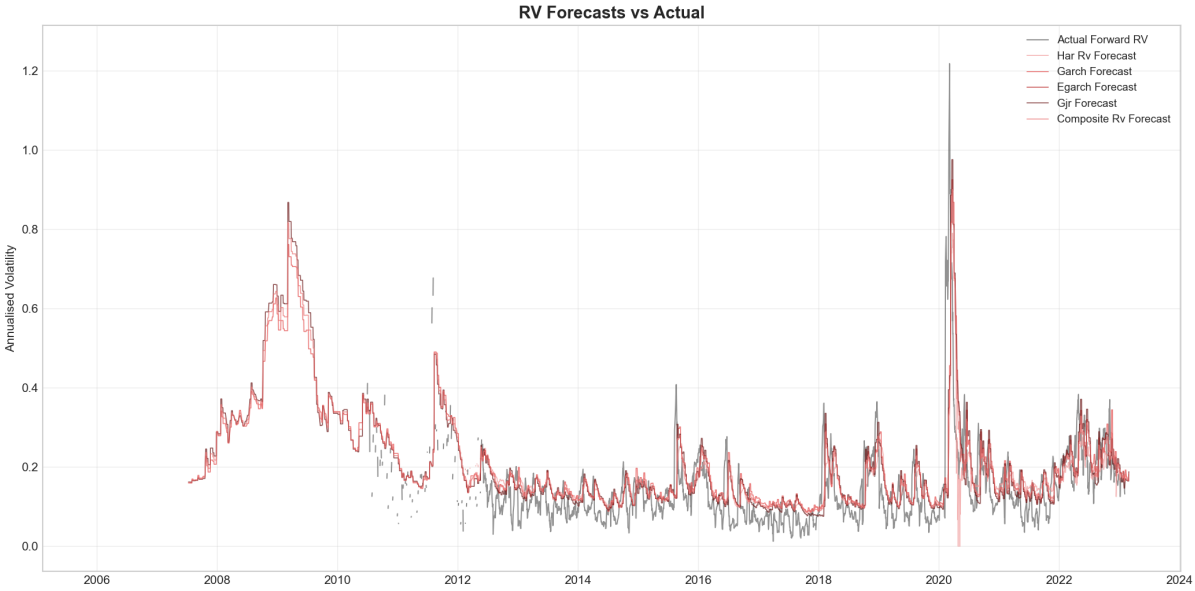

Figure 13: Rolling RV forecast models: HAR-RV, GARCH, GJR-GARCH, and composite.

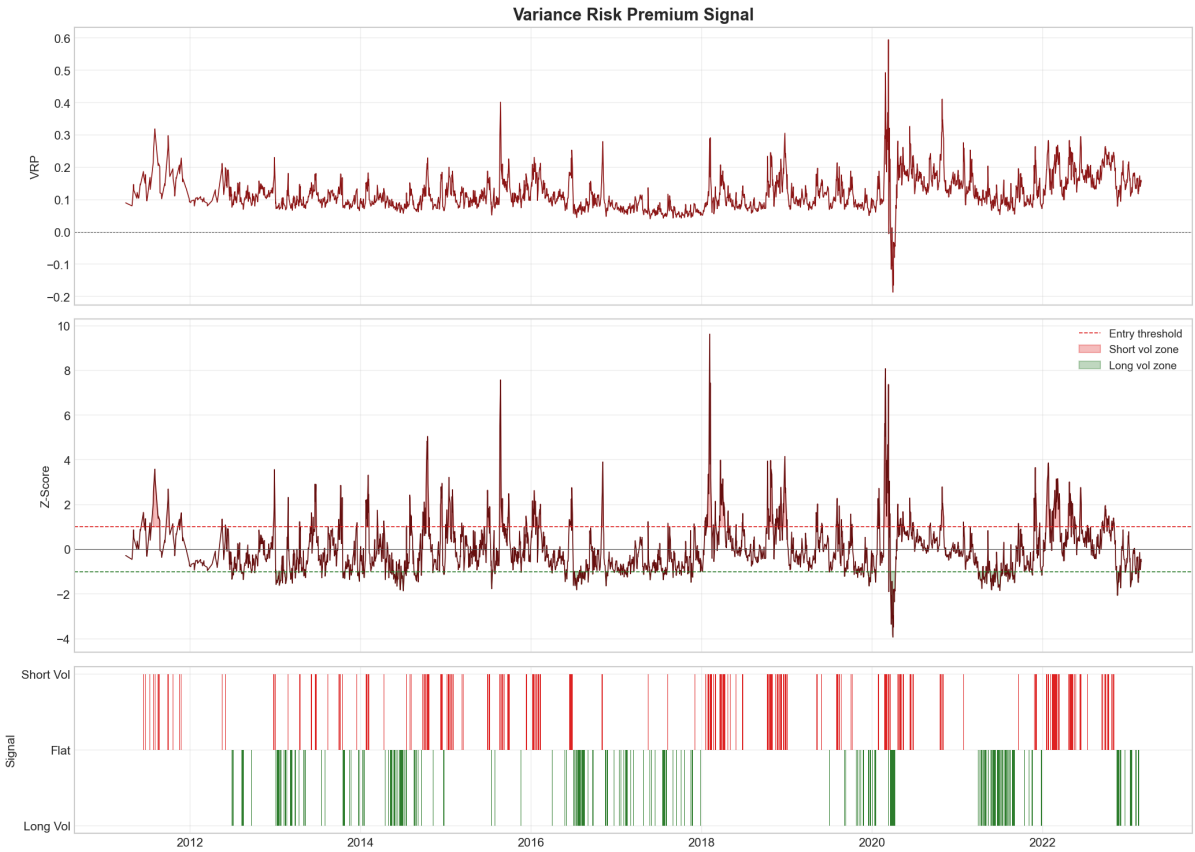

Figure 14: VRP signal z-score and discrete trading signals over time.

Figure 15: Skew and term-structure surface signals.

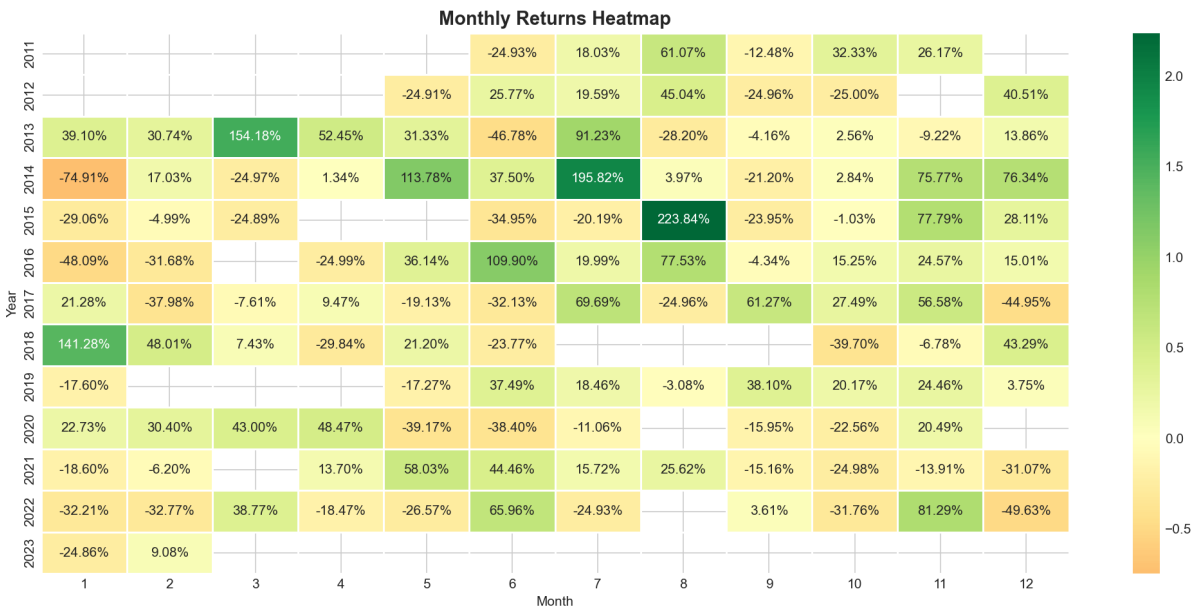

Figure 16: Monthly return heatmap.

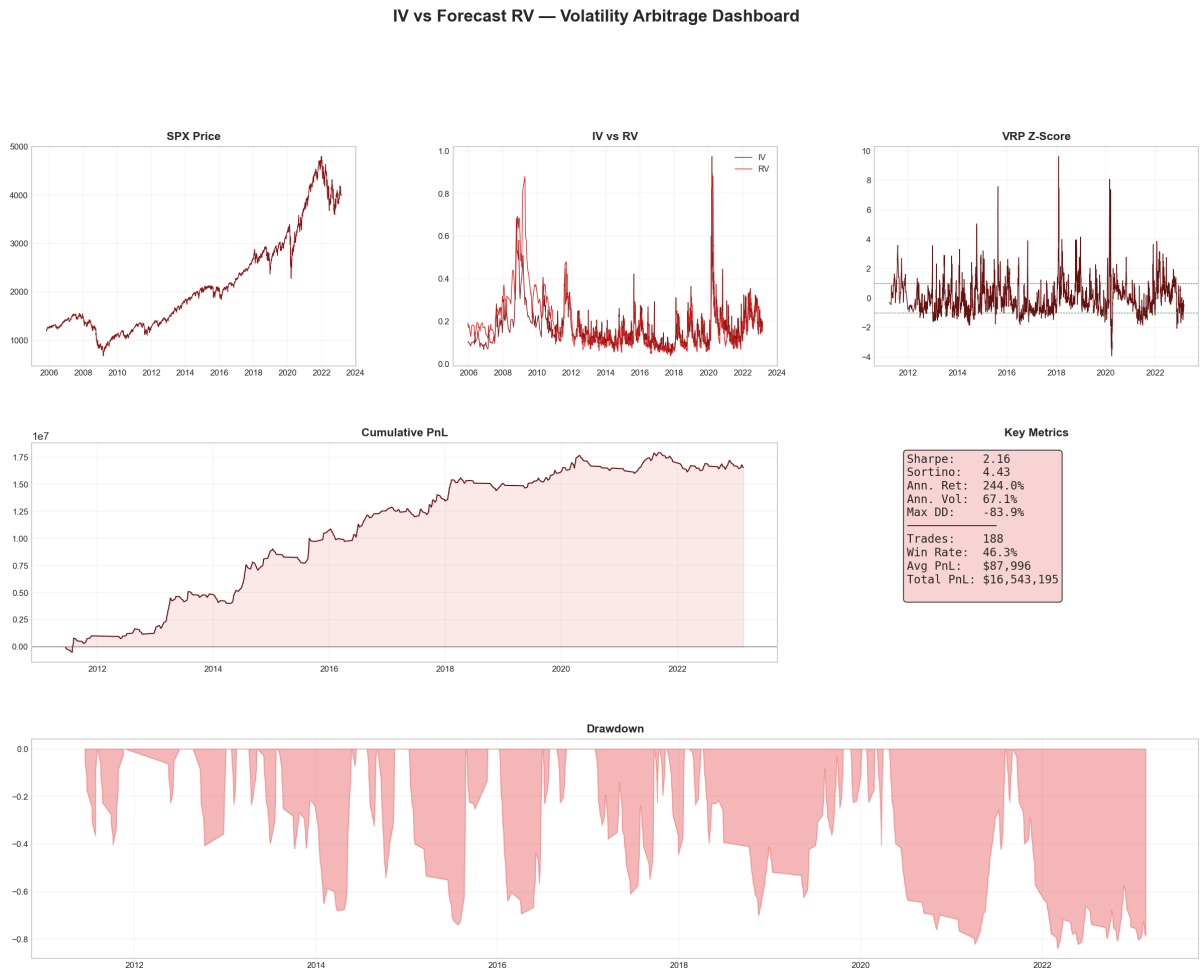

Figure 17: Summary performance dashboard.